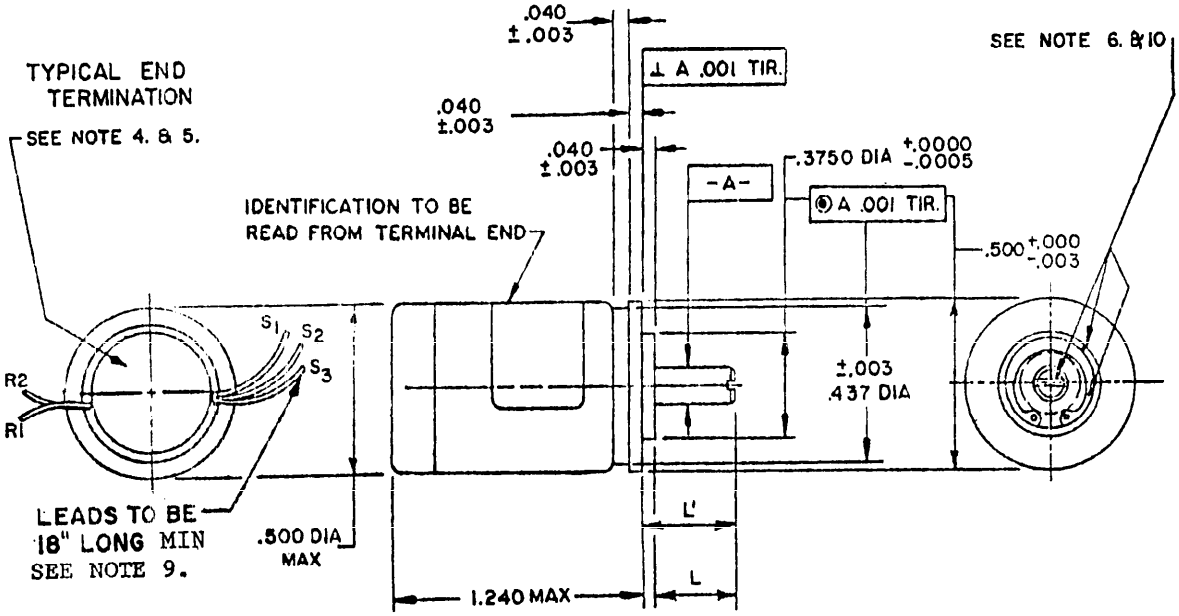
MIL-S-20708/135A
9 July 1971
SUPERSEDING
MIL-S-20708/135
12 September 1966

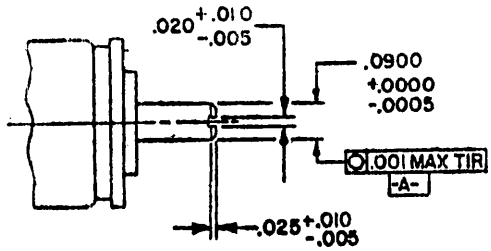
MILITARY SPECIFICATION SHEET

SYNCHRO, TRANSMITTER, TYPE 26V-05CX4A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The Complete requirements for procuring the synchro described herein shall consist of this document and the latest issue of Specification MIL-S-20708.





FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are ± .005 and angles ± 1°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal lead wires shall depart through the back surface of the terminal end of the synchro in any acceptable manner.
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. Mark on housing is to match shaft mark within 10° when rotor is set at approx. synchro zero, and will bear no relation to the center lines shown on the drawing.
7. End play shall be measured as total movement during reversal of 4 oz. axial load.
8. Radial play shall be measured as close to the bearing as possible during reversal of 4 oz. load applied not more than 1/8 inch from end of shaft.
9. Wire leads shall be type ET-30, MIL-W-16878/6.
10. The front retainer shall be threaded or a snap ring.
11. Pending issuance of a qualified products list (QPL) a sample of the synchros to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-S-20708.
12. Endurance shall be performed at 600 rpm.

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SYNCHRO, TRANSMITTER, TYPE 26V-05CX4A

TABLE I PERFORMANCE REQUIREMENTS

Requirement	Value	Unit	Tolerance	Degraded Value	Degraded Value 1/
Frequency	400	Hz	±1%	--------	--------
Primary Voltage	26	volts	±1%	--------	--------
Primary Current	60.0	milliamps	maximum	--------	--------
Primary Power	0.35	watts	nominal	--------	--------
Impedance		ohms	R±20%+jx±20%	--------	--------
Zro	150+j520	--------	---------	--------	--------
Zss	56+j8.6	--------	---------	--------	--------
Transformation Ratio	0.454	--------	±3%	--------	--------
Phase Shift	16 (lead)	degrees	maximum	--------	--------
Electrical Error 5/	10	minutes	maximum	--------	15
Null Voltage:		millivolts	maximum	--------	--------
Total	40	--------	--------	60 2/ 3/	60
Fundamental	30	--------	--------	45 2/ 3/	45
Friction Torque	3	g. cm.	maximum	9 3/ 4/	9
Radial Play	0.0008	inches	maximum	0.0012 3/	0.0012
End Play	0.0001-0.0010	inches	min.-max.	0.0001-0.0015 3/	0.0001-0.0015
Brush Contact Res.	Meas. D.C. Rotor Res.	ohms	1 or 0.5%	--------	1.5 or 0.75%
Dielectric Withstand.	1.0 (peak)	milliamps	maximum	--------	1.5 (peak)
Insulation Res.	50	megohms	minimum	10 2/	25
Wire Lead Stress 6/	1	pound	maximum	--------	--------
Temperature Rise	10	°C	maximum	--------	--------

NOTES:
1/ Applicable after High Impact Shock
2/ Applicable during High Temperature
3/ Applicable after Endurance
4/ Applicable during Low Temperature
5/ An additional tolerance of ±3 minutes is allowed to compensate for mounting type eccentricity errors for Size 05 Synchros encountered during testing
6/ Wire lead stress shall be performed as a qualification test

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TABLE II. Dash Number and Characteristics

Dash No.	Shaft diameter	Shaft style	Termination	Shaft length	
				L ref	L' dim to first mounting flange surface (±.010)
-001	.0900	Plain	Wire leads	.335	.375

Part Number: M20708/135A and a dash number from Table II.

Qualification - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Frankford Arsenal, Philadelphia, Pa. 19137, ATTN: SMUFA-T1000 and information pertaining to qualification of products may be obtained from that activity.

Custodians:
Army - MU
Navy - AS
Air Force - 80

Preparing activity:
Army - MU

Project No. 5990-0277

Review activities:
Army - EL, ME, MI, SL
Navy - AS
Air Force - 17, 11
DSA - ES

User activities:
Army - AV, WC, AT
Navy - MC, OS
Air Force - None

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Department of the Army

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Commanding Officer
Frankford Arsenal
ATTN: SMUFA-T1000
Philadelphia, Pa. 19137

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